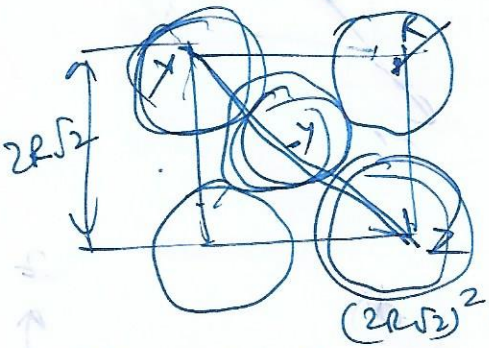
Tutorial (1)

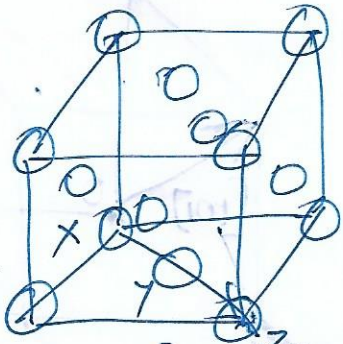
10/8/23

Linear Density (LD) → no. of atoms per unit length whose centers lie on the direction vector for a specific crystallographic dirn.

$$LD = \frac{\text{no. of atoms centered on direction vector}}{\text{length of direction vector}}$$

$$LD_{110} = \frac{2 \text{ atoms}}{4R} \quad (\rightarrow \tfrac{1}{2} \times 2 + 1) = \frac{1}{2R}$$





unit → nm^{-1} or m^{-1} etc.

$(2R\sqrt{2})^2 + (2R\sqrt{2})^2 = 8R^2 + 8R^2 = 16R^2 = (4R)^2$

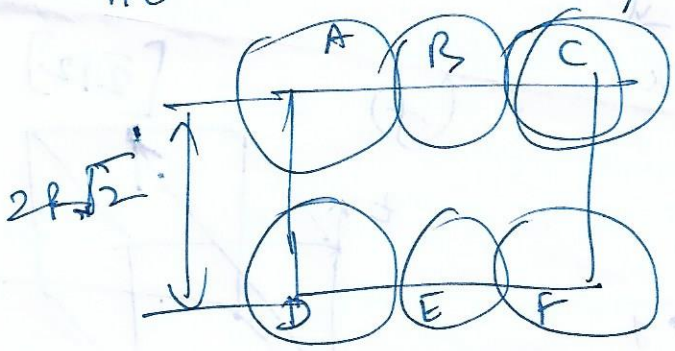
Planar Density (PD) → no. of atoms / unit area that are centered on a particular crystallogr. plane

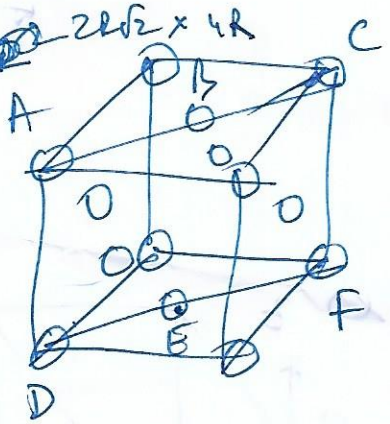
$$PD = \frac{\text{no. of atoms centered on a plane}}{\text{area of plane}}$$

unit → nm^{-2}, m^{-2} etc.

A, C, D, F ← $\frac{1}{4} \times 4 + \frac{1}{2} \times 2$ → B, E

$$PD_{110} = \frac{2}{2R \times 4R\sqrt{2}}$$





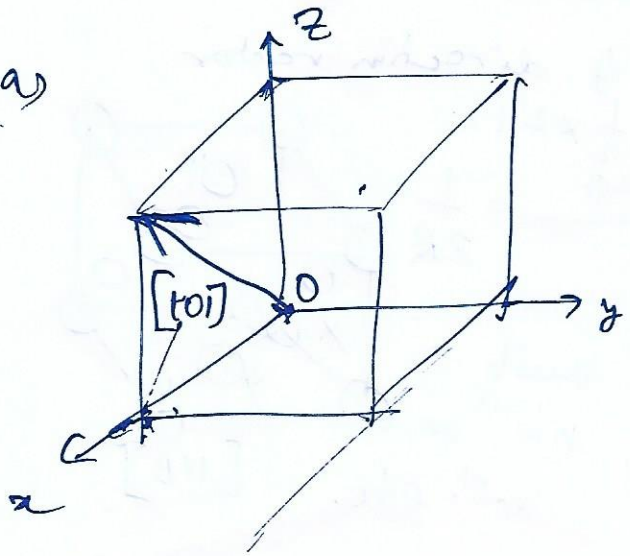
$$= \frac{2}{8R^2\sqrt{2}} = \frac{1}{4R^2\sqrt{2}}$$

(2) Within a cubic unit cell, sketch the following directions:

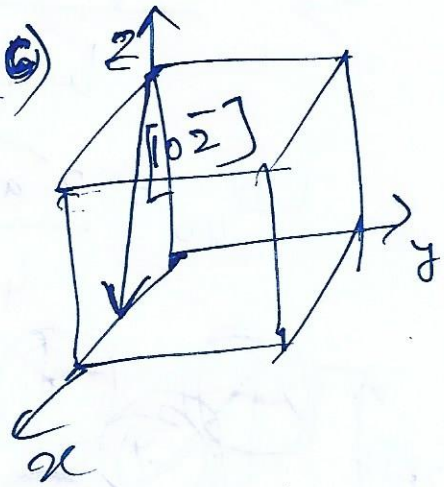
(a) $[101]$ (b) $[211]$ (c) $[10\bar{2}]$ (d) $[3\bar{1}3]$

(e) $[\bar{1}1\bar{1}]$ (f) $[\bar{2}12]$ (g) $[3\bar{1}2]$ (h) $[301]$

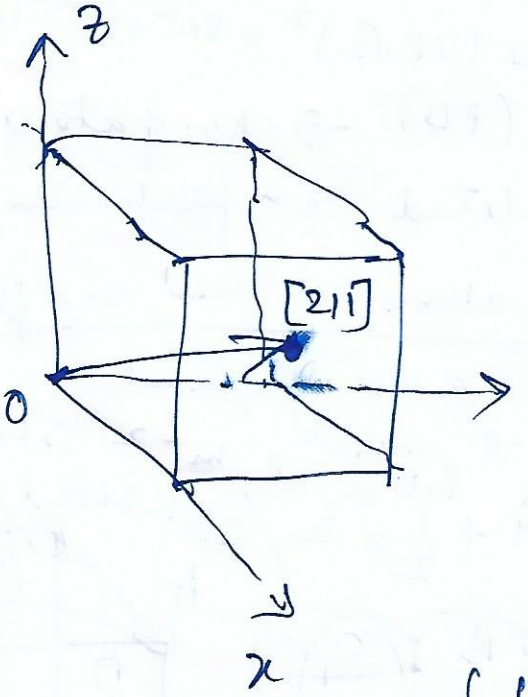
(a)



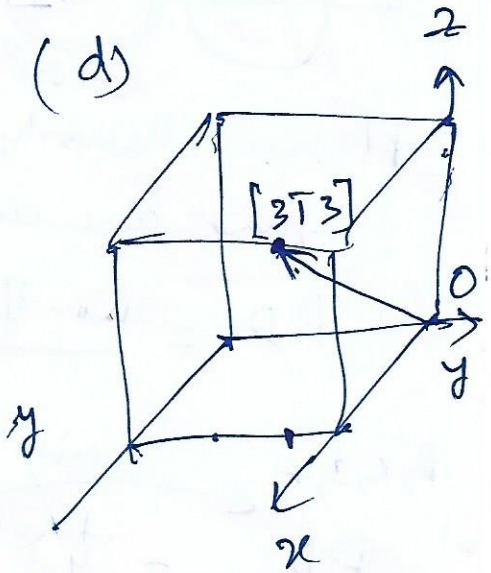
(c)



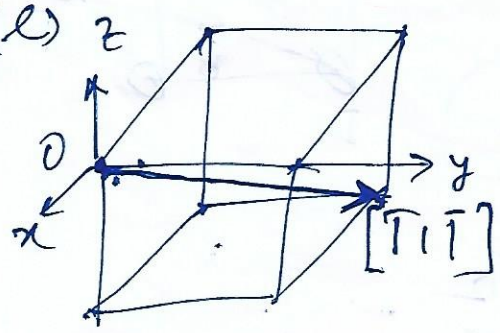
(b)



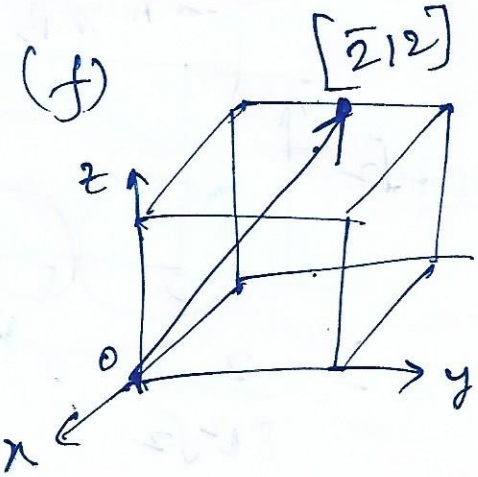
(d)



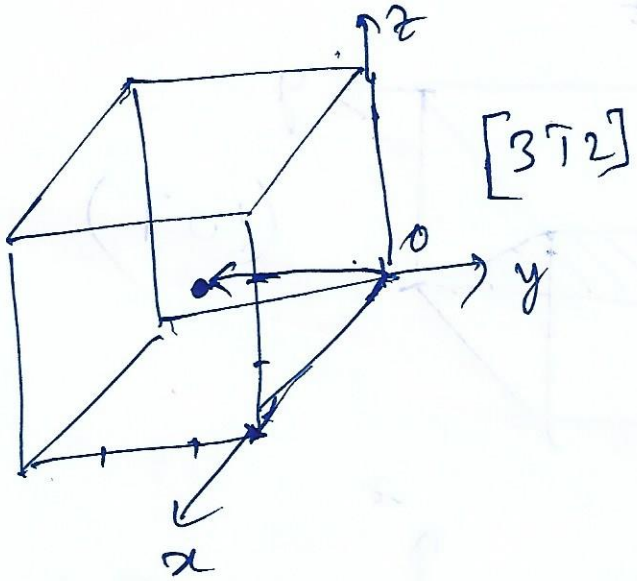
(e)



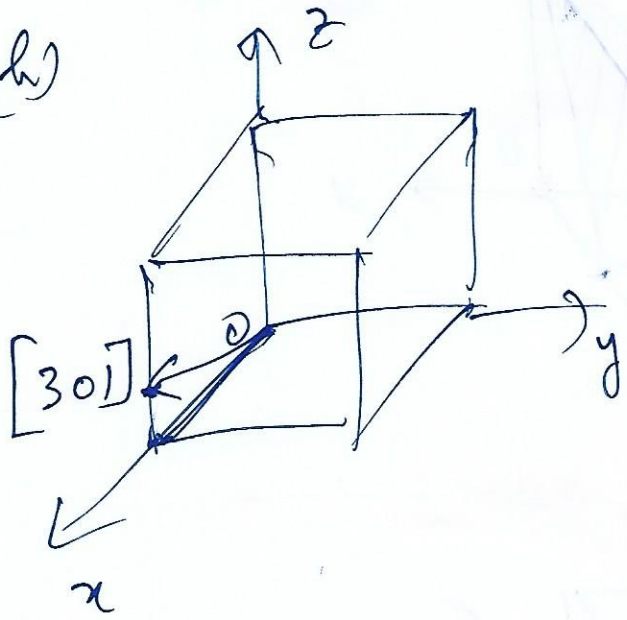
(f)



(g)



(h)



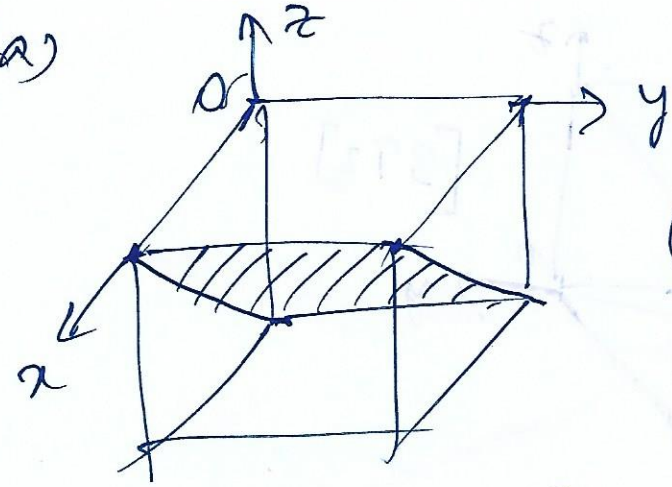
(3) Sketch within a cubic unit cell the following planes

(a) (1 0 $\bar{1}$) (b) (2 $\bar{1}$ 1) (c) (0 1 2)

(d) (3 $\bar{1}$ 3) (e) ($\bar{1}$ 1 $\bar{1}$) (f) ($\bar{2}$ 1 2)

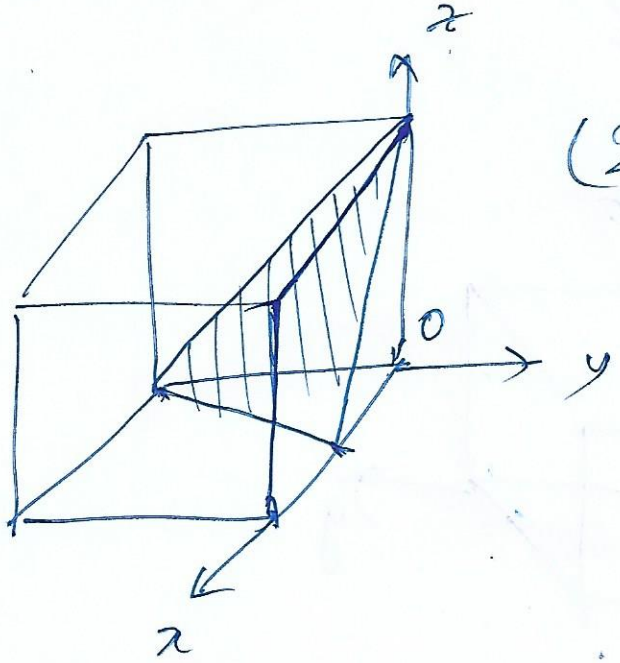
(g) (3 $\bar{1}$ 2) (h) (3 0 1)

(a)



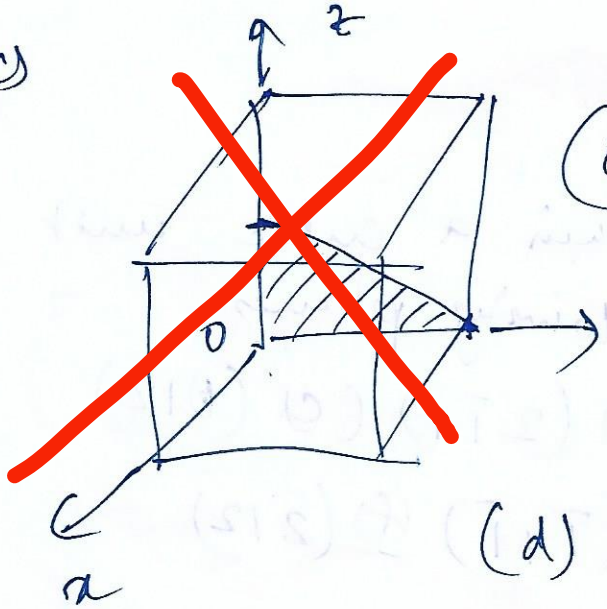
(10T)

(b)



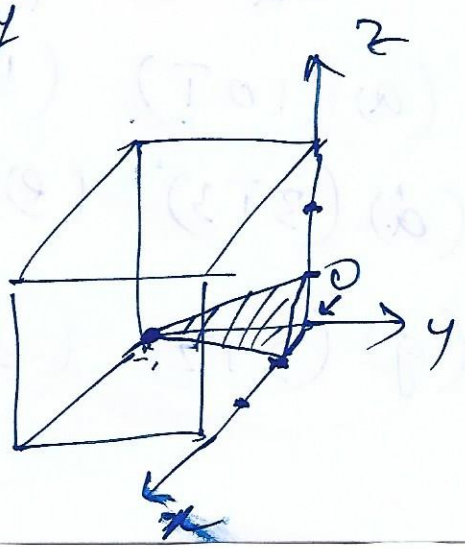
(2T1)

(c)



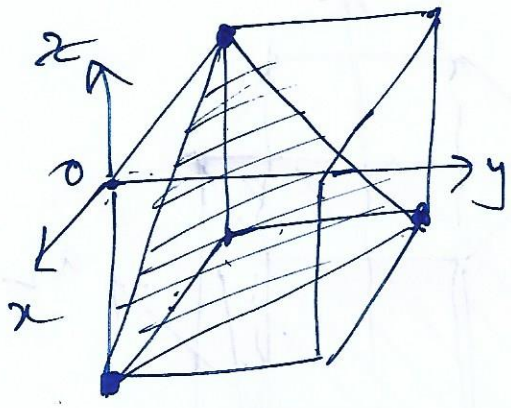
(012) See last page

(d)

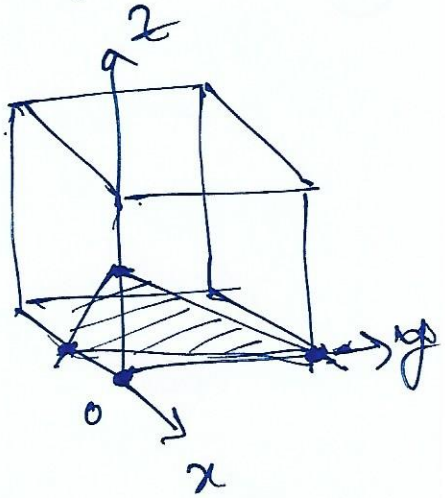


(3T3)

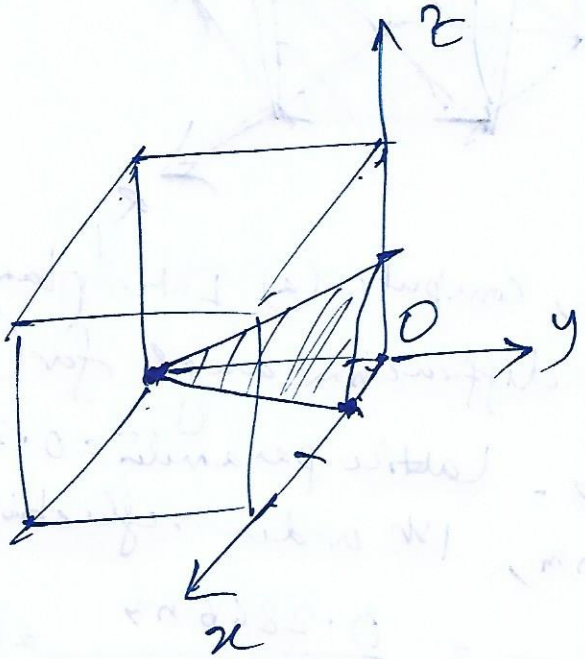
(e) $(\bar{1}\,1\,\bar{1})$



(f) $(\bar{2}\,1\,2)$

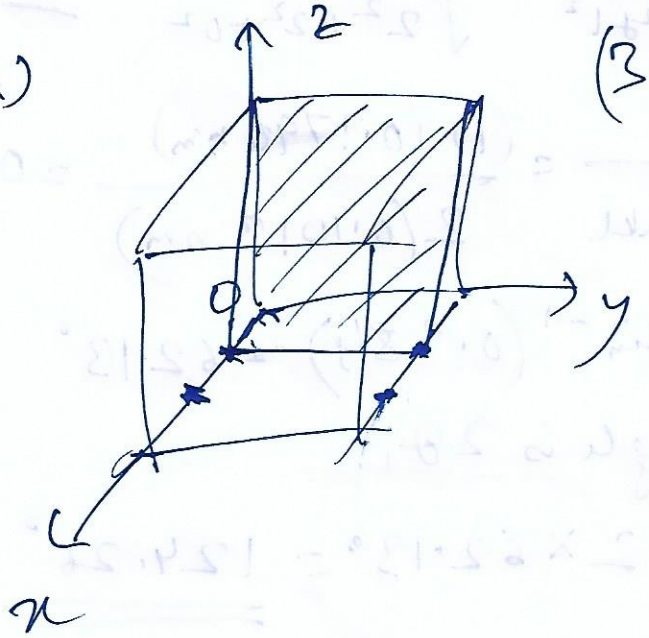


(g)



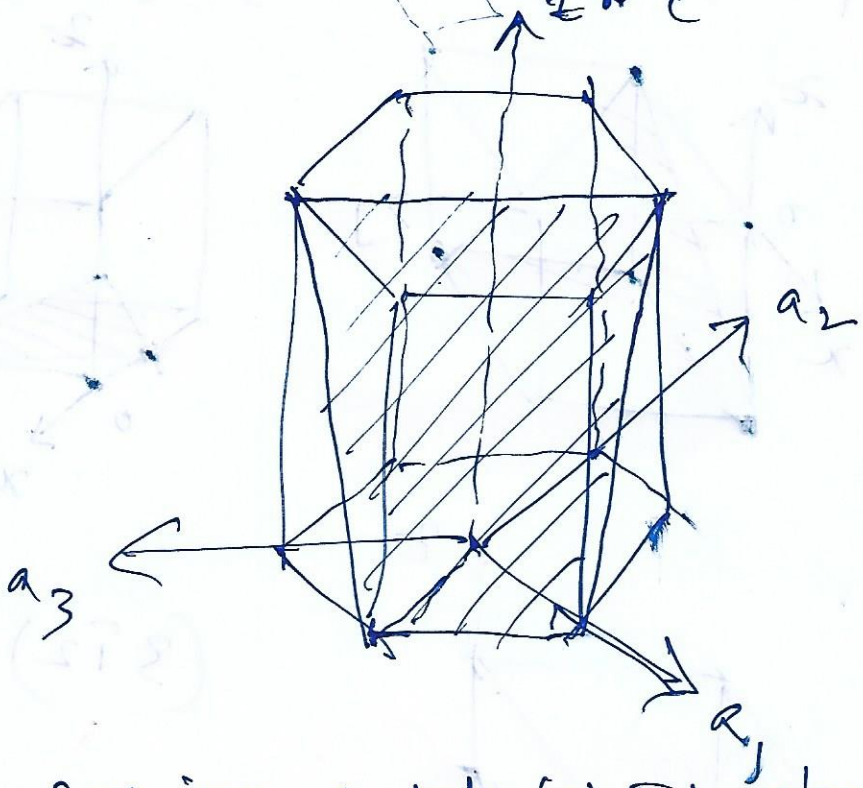
$(3\,\bar{1}\,2)$

(h)



$(3\,0\,1)$

(4) Draw (1 $\bar{1}$ 0 1) in Hexagonal unit cell



(5) For BCC iron, compute (a) Interplanar spacing & (b) diffraction angle for (220) set of planes. Lattice parameter = 0.2866 nm. λ = 0.1790 nm, 1st order reflection.

(a) $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}} = \frac{0.2866\ nm}{\sqrt{2^2+2^2+0^2}} = 0.1013\ nm$

(b) $\sin\theta = \frac{n\lambda}{2d_{hkl}} = \frac{(1)(0.1790\ nm)}{2(0.1013\ nm)} = 0.884$

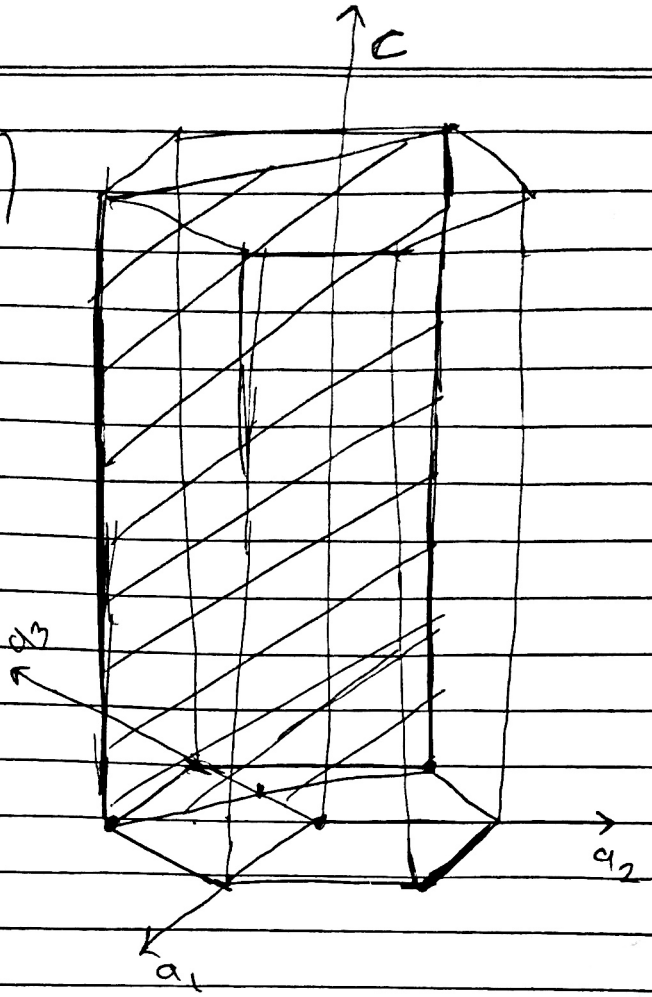
$\theta = \sin^{-1}(0.884) = 62.13^\circ$

Diffraction angle is 2θ

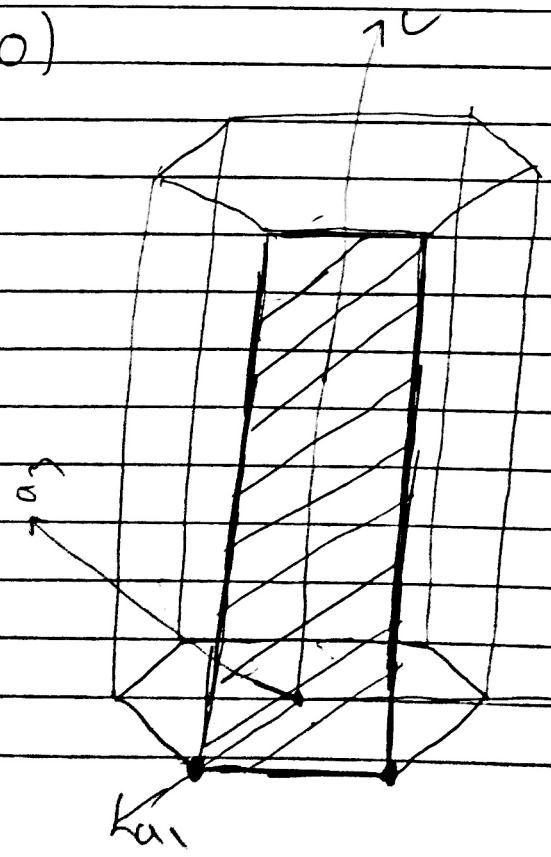
$\therefore 2\theta = 2 \times 62.13^\circ = 124.26^\circ$

Q.(4)

(ii) $(\bar{1}\bar{1}20)$



(iii) $(10\bar{1}0)$



3 (c) (0,1,2) (Corrected)

